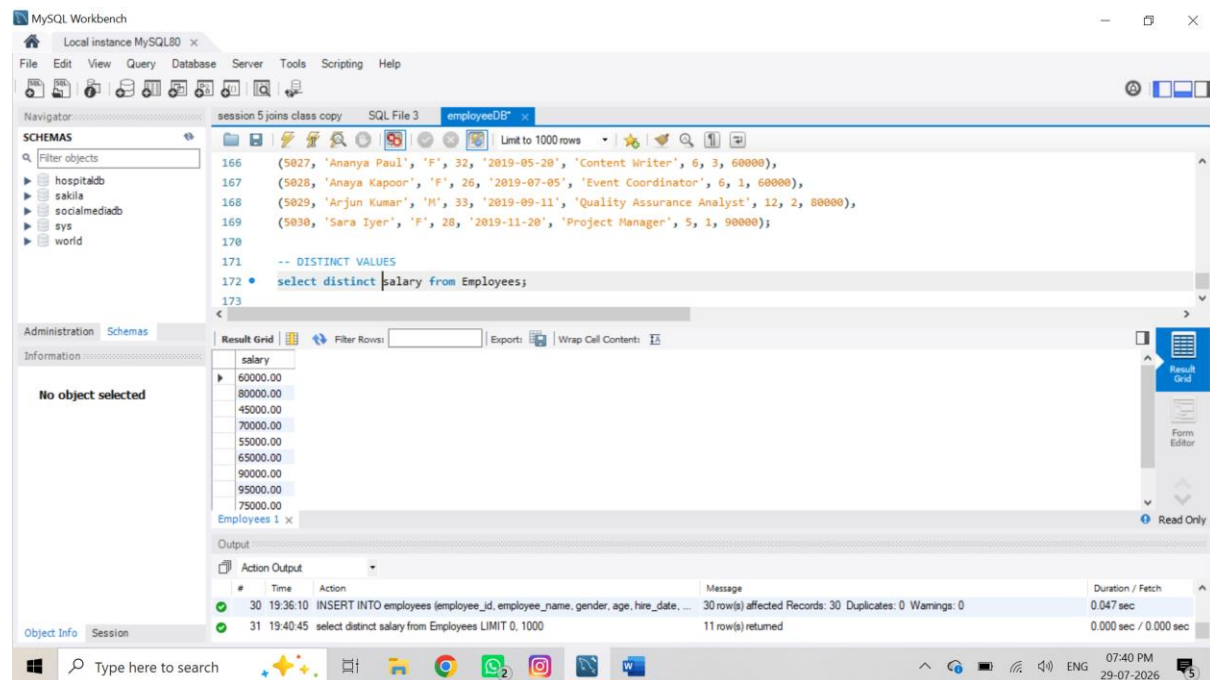


Clause & Operators:

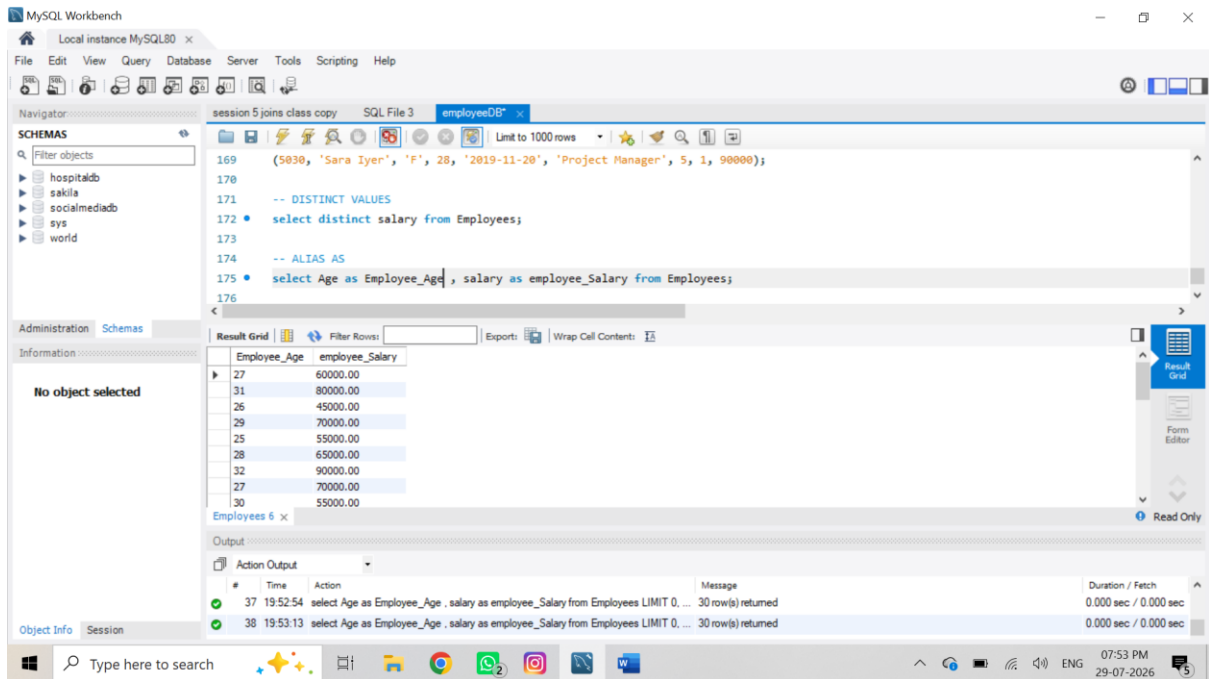
1. DISTINCT VALUES:

- A query to retrieve distinct salaries from the Employees table.



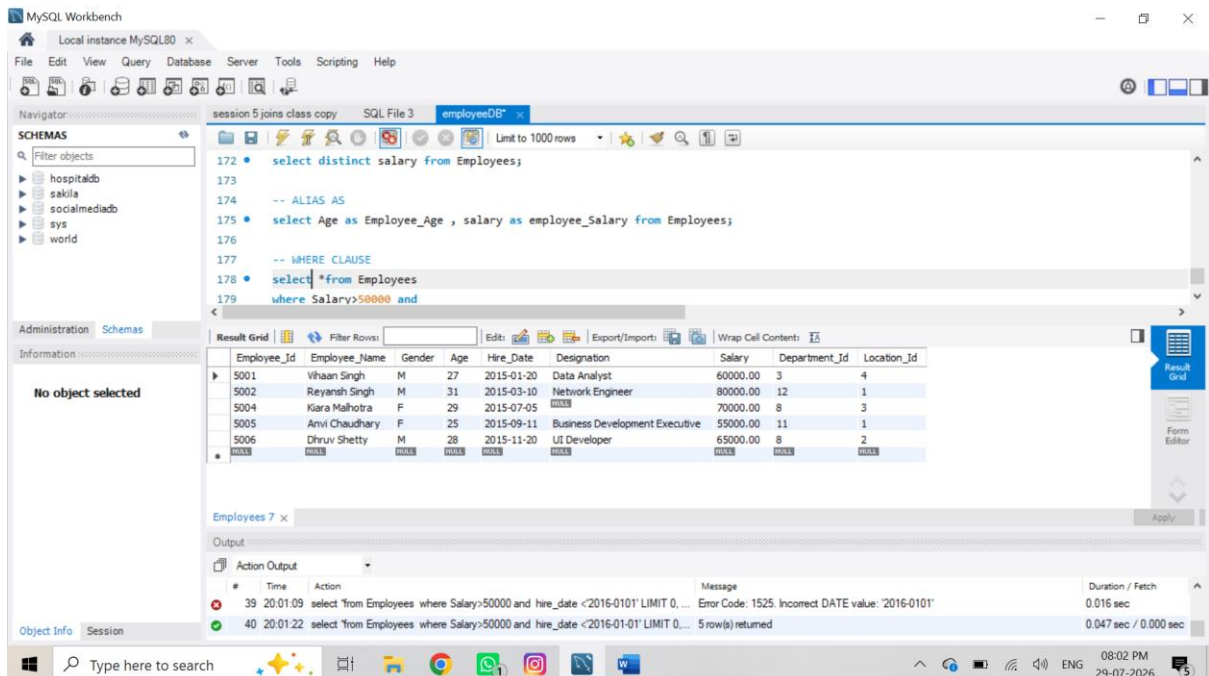
2. ALIAS (AS):

- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

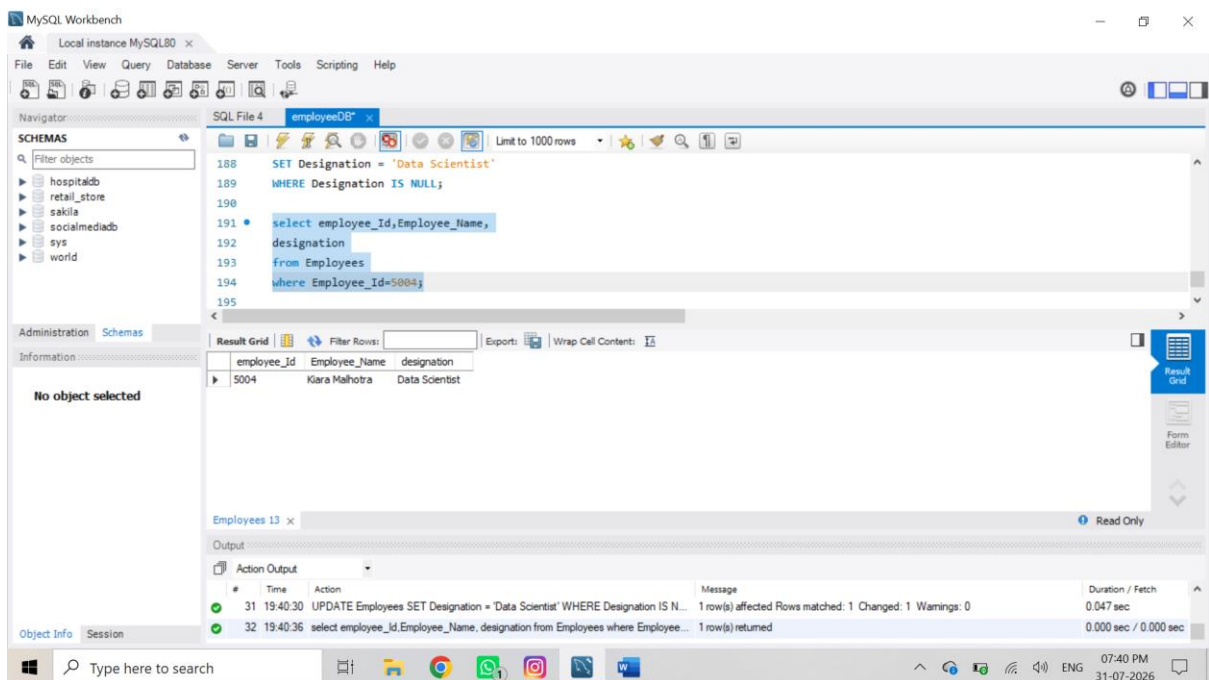
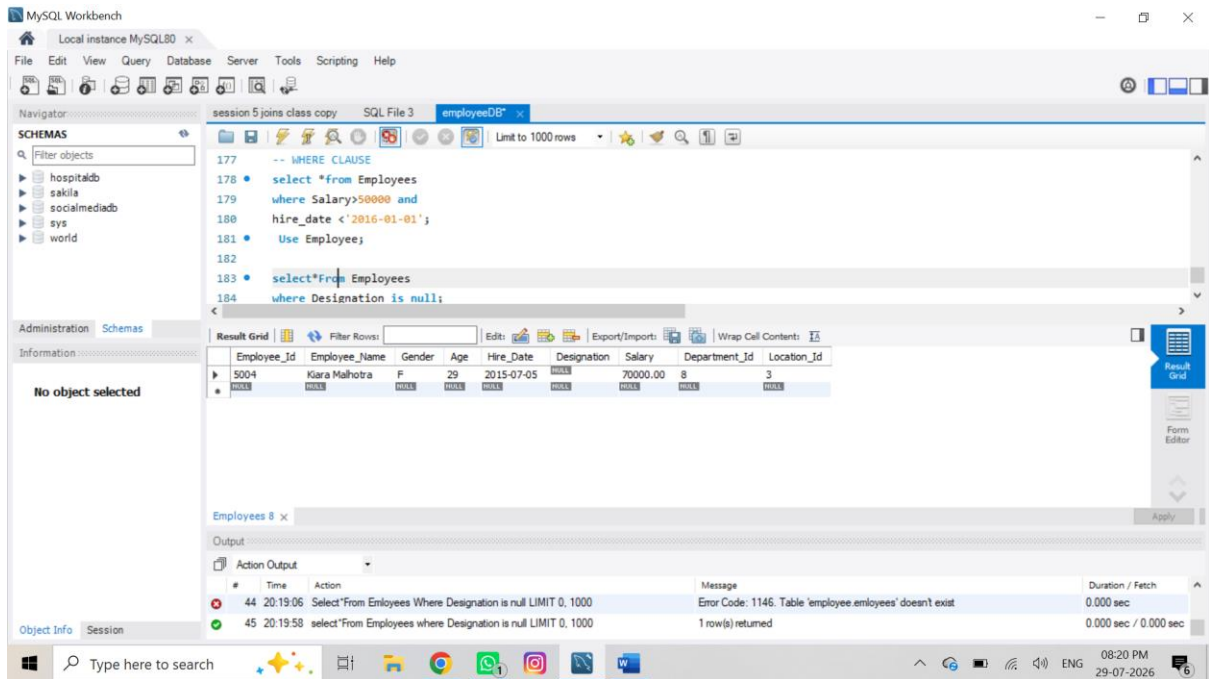


3. WHERE CLAUSE & OPERATORS:

- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.



- Find the employee whose designation is missing and fill it with "Data Scientist".



Sorting and Grouping Data:

ORDER BY:

- Find employees sorted by department ID in ascending order and salary in descending order.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

hospitaldb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

Information

No object selected

SQL File 4 employeeDB x

```

193 from Employees
194 where Employee_Id=5004;
195
196 -- Find employees sorted by department ID in ascending order and salary indescending order
197
198 • select* from Employees
199 order by Department_Id ASC,Salary DESC;
200

```

Result Grid

Employee_Id	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_Id	Location_Id
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	2	1
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3	4
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	75000.00	3	2
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	4	4
5022	Aadhya Ivenoor	F	28	2018-07-10	HR Specialist	60000.00	4	4

Employees 14 x

Output

Action Output

Time Action Message Duration / Fetch

33 19:45:54 select* from Employees order by Department_Id ASC,Salary DESC LIMIT 0, 1000 Error Code: 1054. Unknown column 'Department_Id' in 'order clause' 0.000 sec

34 19:48:28 select* from Employees order by Department_Id ASC,Salary DESC LIMIT 0, 1000 30 row(s) returned 0.015 sec / 0.000 sec

Object Info Session

Type here to search

07:49 PM 31-07-2026

2. LIMIT:

- Display the first 5 employees hired in the year 2018.

MySQL Workbench

Local instance MySQL80 x

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Navigator

SCHEMAS

Filter objects

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retail_store
sakila
socialmediadb
sys
world

Administration Schemas

Information

No object selected

SQL File 4 employeeDB x

```

199 order by Department_Id ASC,Salary DESC;
200
201 -- Display the first 5 employees hired in the year 2018
202
203 • select * from Employees
204 where Year(Hire_Date)=2018
205 order by Hire_date ASC limit 5;
206

```

Result Grid

Employee_Id	Employee_Name	Gender	Age	Hire_Date	Designation	Salary	Department_Id	Location_Id
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2

Employees 15 x

Output

Action Output

Time Action Message Duration / Fetch

34 19:48:28 select* from Employees order by Department_Id ASC,Salary DESC LIMIT 0, 1000 30 row(s) returned 0.015 sec / 0.000 sec

35 19:53:40 select * from Employees where Year(Hire_Date)=2018 order by Hire_date ASC limit 5 5 row(s) returned 0.000 sec / 0.000 sec

Object Info Session

Type here to search

07:53 PM 31-07-2026

3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

208
209 • SELECT SUM(e.Salary) AS Total_Finance_Salary
210 FROM Employees e
211 JOIN Departments d
212 ON e.Department_Id = d.Department_Id
213 WHERE d.Department_Name = 'Finance';
214
215

```

The Result Grid shows the output of the query:

Total_Finance_Salary
170000.00

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
38	20:22:31	select sum(salary) as Total_Finance_Salary from Employees where Department_Id=7...	1 row(s) returned	0.000 sec / 0.000 sec
39	20:24:01	SELECT SUM(e.Salary) AS Total_Finance_Salary FROM Employees e JOIN Depart...	1 row(s) returned	0.000 sec / 0.000 sec

- Find the minimum age among all employees.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

215
216 -- Find the minimum age among all employees
217
218 • Select Employee_Id, Employee_Name, Age
219 from Employees
220 where Age=(Select MIN(Age) from
221 Employees);
222

```

The Result Grid shows the output of the query:

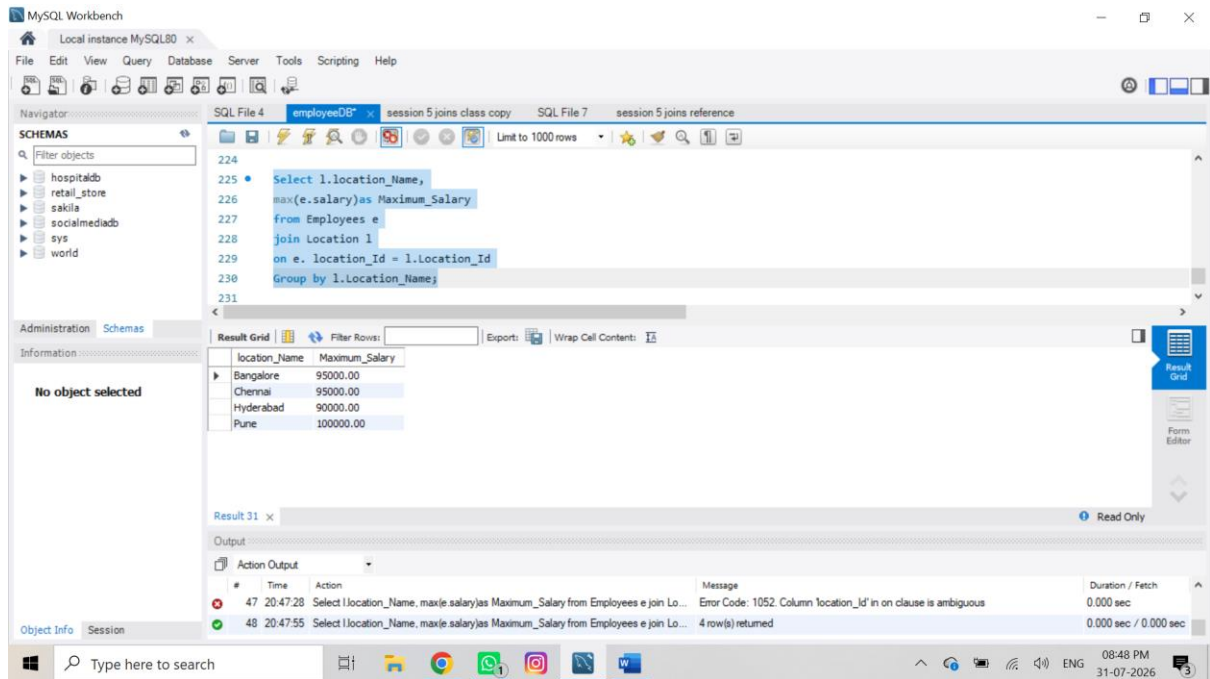
Employee_Id	Employee_Name	Age
5005	Anvi Chaudhary	25
5018	Ishan Mishra	25

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
42	20:37:33	Select Employee_Id, Employee_Name, Age from Employees Where Age=(Select MI...	2 row(s) returned	0.000 sec / 0.000 sec
43	20:38:36	Select Employee_Id, Employee_Name, Age from Employees Where Age=(Select MI...	2 row(s) returned	0.000 sec / 0.000 sec

4. GROUP BY:

- List the maximum salary for each location.



- Calculate the average salary for each designation containing the word 'Analyst'.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

hospitadb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

No object selected

SQL File 4 employeeDB session 5 joins class copy SQL File 7 session 5 joins reference

Limit to 1000 rows

```

231
232 -- Calculate the average salary for each designation containing the word Analyst
233 • Select Designation,
234 Avg(salary) As Average_Salary
235 from Employees
236 where Designation like '%Analyst%'
237 Group by Designation;
238

```

Result Grid

Designation	Average_Salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

Result 32

Output

Action Output

#	Time	Action	Message	Duration / Fetch
48	20:47:55	Select Location_Name, max(salary) as Maximum_Salary from Employees e join Lo...	4 row(s) returned	0.000 sec / 0.000 sec
49	20:53:10	Select Designation, Avg(salary) As Average_Salary from Employees where Designat...	7 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

08:53 PM 31-07-2026

5. HAVING:

- Find departments with less than 3 employees.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

hospitadb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

No object selected

SQL File 4 employeeDB session 5 joins class copy SQL File 7 session 5 joins reference

Limit to 1000 rows

```

239 -- Find departments with less than 3 employees
240
241 • Select department_Id,
242 Count(Employee_Id) As Total_Employees
243 From Employees
244 Group by Department_Id
245 having Count(Employee_Id)<3;
246

```

Result Grid

department_Id	Total_Employees
1	1
5	2
6	2
7	2
9	1
11	2
13	1

Result 33

Output

Action Output

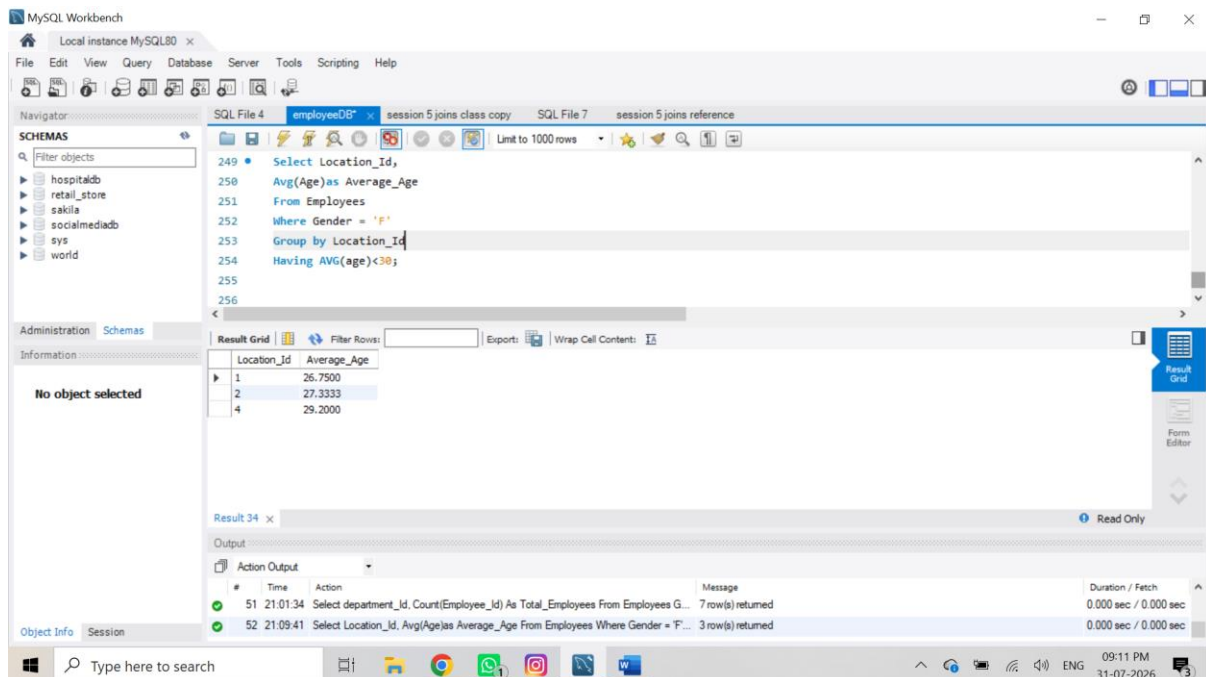
#	Time	Action	Message	Duration / Fetch
50	20:59:41	Select department_Id, Count(Employee_Id) As Total_Employees From Employees ...	Error Code: 1054, Unknown column 'Employee_Id' in field list	0.000 sec
51	21:01:34	Select department_Id, Count(Employee_Id) As Total_Employees From Employees G...	7 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

09:01 PM 31-07-2026

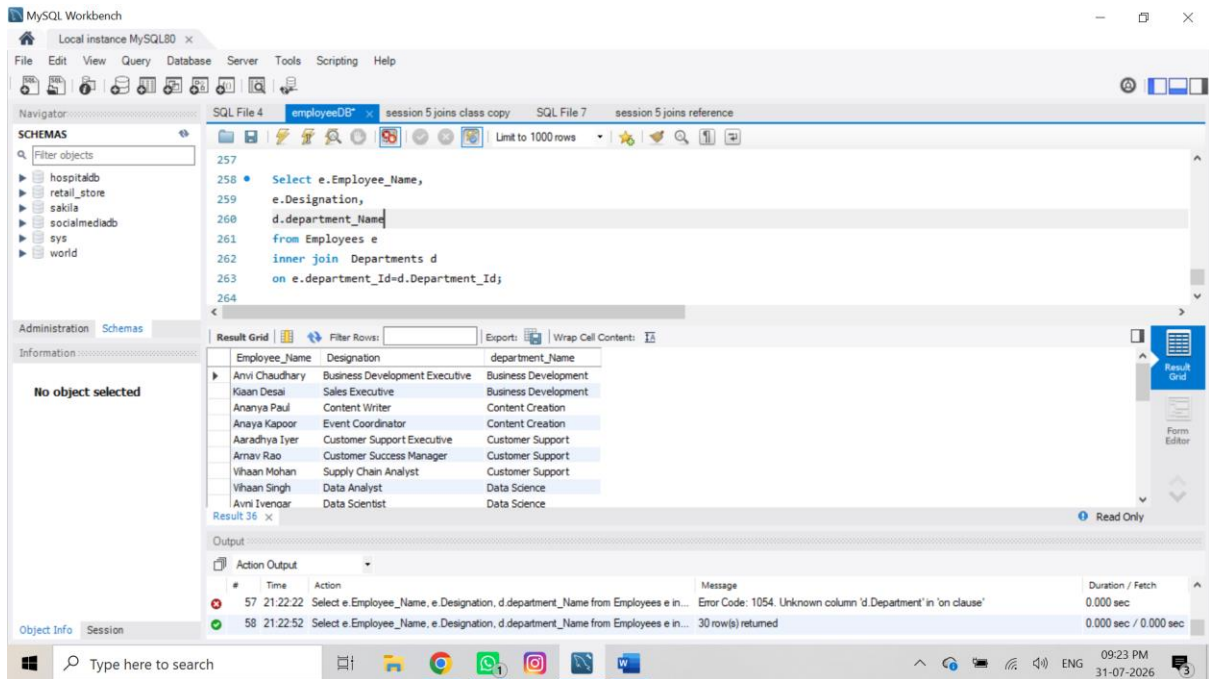
- Find locations with female employees whose average age is below 30.



Joins:

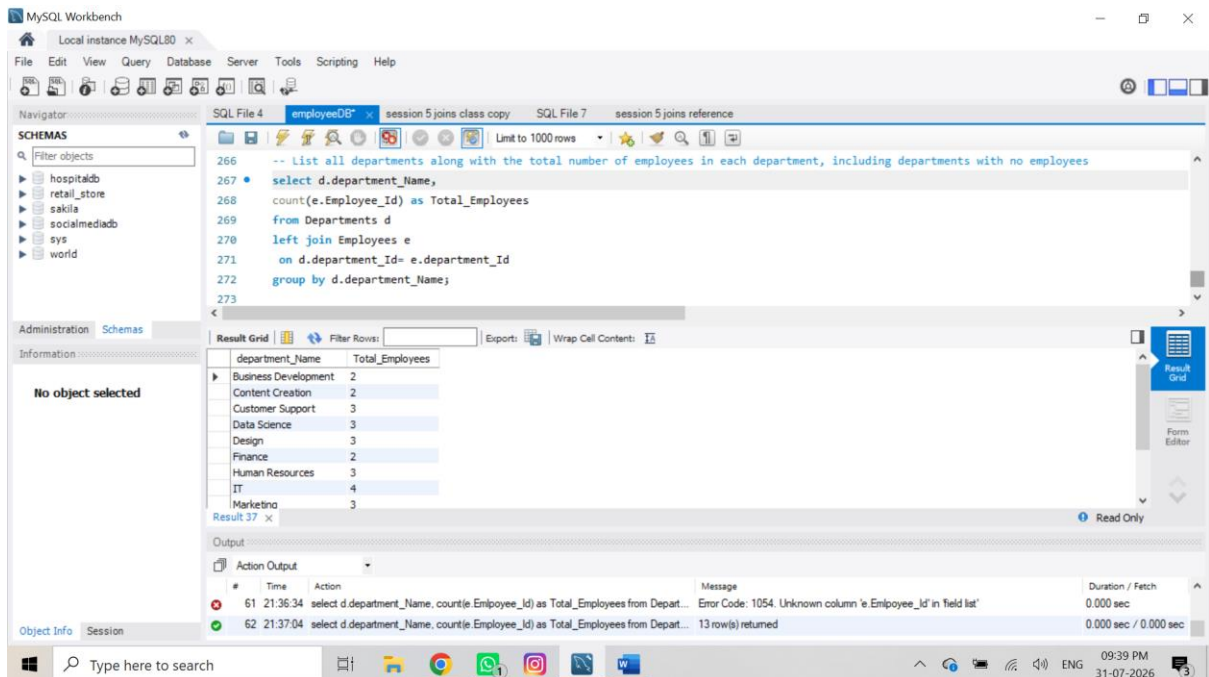
1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.



2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.



3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location.

If no employees are assigned to a location, display NULL for employee name.

The screenshot shows the MySQL Workbench interface with a SQL query in the editor. The query is as follows:

```

275
276 • Select
277   l.Location_Name,
278   e.Employee_Name
279 From Employees e
280 Right join Location l
281 on e.location_Id=l.location_Id;
282

```

The result grid shows the following data:

Location_Name	Employee_Name
Bangalore	Aaradhyas Iyer
Bangalore	Dhruv Shetty
Bangalore	Myra Verma
Bangalore	Vihan Mohan
Bangalore	Kabir Nair
Bangalore	Anika Paul
Bangalore	Vivaan Singh
Bangalore	Arjun Kumar
Chennai	Revansh Singh

The output pane shows the execution of the query, indicating that 30 rows were returned.

4. CROSS JOIN

- Show all possible combinations of departments and locations.

The screenshot shows the MySQL Workbench interface with a SQL query in the editor. The query is as follows:

```

282 • use employee;
283 -- Show all possible combinations of departments and locations
284 • select
285   d.Department_Name,
286   l.location_Name
287 from departments d
288 cross join location l;
289

```

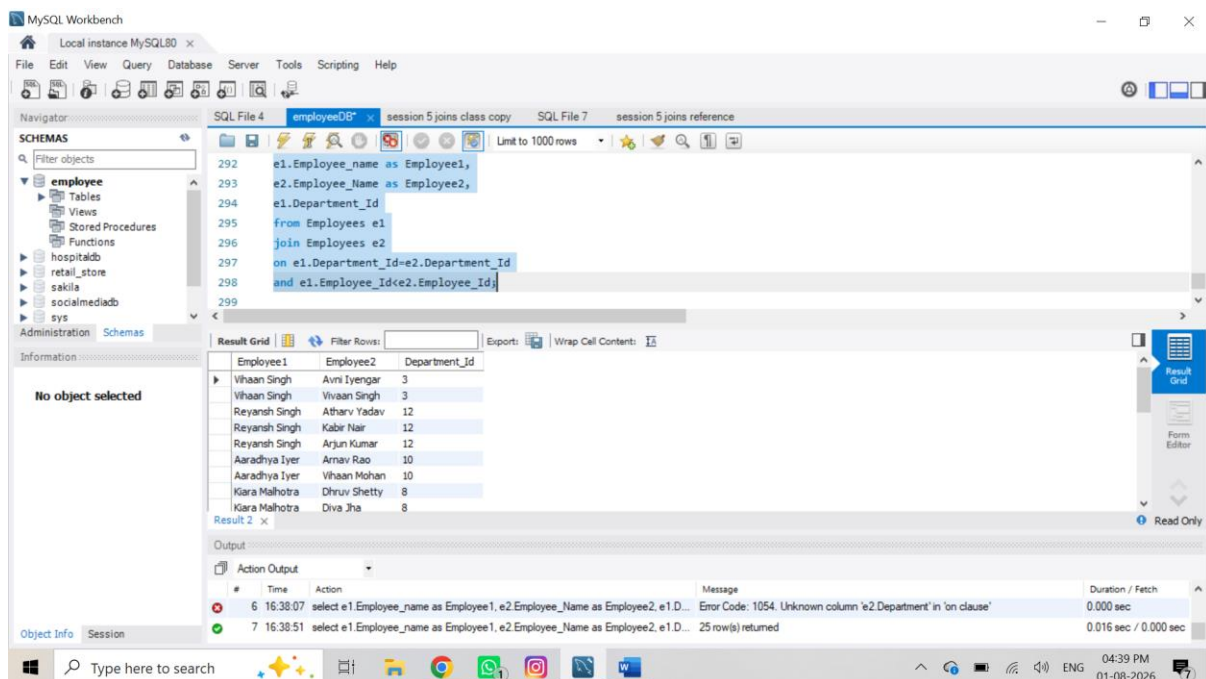
The result grid shows the following data:

Department_Name	location_Name
Business Development	Pune
Business Development	Hyderabad
Business Development	Chennai
Business Development	Bangalore
Content Creation	Pune
Content Creation	Hyderabad
Content Creation	Chennai
Content Creation	Bangalore
Customer Support	Pune

The output pane shows the execution of the query, indicating that 52 rows were returned.

5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.



Windows function

- Write a window function query to rank employees by salary using rank().

MySQL Workbench

Local instance MySQL80

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Filter objects

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Functions

hospitaldb

retail_store

sakila

socialmediadb

sys

Administration Schemas

Information

No object selected

SQL File 4 employeeDB session 5 joins class copy SQL File 7 session 5 joins reference

Limit to 1000 rows

```

301 select
302 Employee_Id,
303 Employee_Name,
304 Salary,
305 Rank() over (order by Salary desc)
306 as Salary_Rank
307 from Employees;
308

```

Result Grid

Employee_Id	Employee_Name	Salary	Salary_Rank
5016	Zoya Khan	100000.00	1
5025	Avni Iyengar	100000.00	1
5012	Myra Verma	95000.00	3
5024	Aryan Shetty	95000.00	3
5007	Anushka Singh	90000.00	5
5020	Aarav Nair	90000.00	5
5030	Sara Iyer	90000.00	5
5015	Ishaan Kumar	85000.00	8
5021	Advik Kacoor	85000.00	8

Result 3

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	16:38:51	select e1.Employee_name as Employee1, e2.Employee_Name as Employee2, e1.D...	25 row(s) returned	0.016 sec / 0.000 sec
8	16:46:34	select Employee_Id, Employee_Name, Salary, Rank() over (order by Salary desc) a...	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

04:46 PM 01-08-2026

- Write a window function query to rank employees by salary within each department using DENSE_RANK()

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

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SCHEMAS

Filter objects

employee

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sakila

socialmediadb

sys

Administration Schemas

Information

No object selected

SQL File 4 employeeDB session 5 joins class copy SQL File 7 session 5 joins reference

Limit to 1000 rows

```

311 Employee_Id,
312 Employee_Name,
313 Department_ID,
314 Salary,
315 dense_Rank() over(partition by department_ID order by Salary DESC)
316 as Salary_Rank
317 from Employees;
318

```

Result Grid

Employee_Id	Employee_Name	Department_ID	Salary	Salary_Rank
5020	Aarav Nair	1	90000.00	1
5007	Anushka Singh	2	90000.00	1
5023	Anika Paul	2	70000.00	2
5011	Saanvi Patel	2	60000.00	3
5025	Avni Iyengar	3	100000.00	1
5026	Vivaan Singh	3	75000.00	2
5001	Vihaan Singh	3	60000.00	3
5016	Zoya Khan	4	100000.00	1
5022	Aadhya Ivenoor	4	60000.00	2

Result 4

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	16:46:34	select Employee_Id, Employee_Name, Salary, Rank() over (order by Salary desc) a...	30 row(s) returned	0.000 sec / 0.000 sec
9	16:52:51	Select Employee_Id, Employee_Name, Department_ID, Salary, dense_Rank() over(...	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

04:53 PM 01-08-2026

- Write a window function query, Running total salary by department

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

Tables

Views

Stored Procedures

Functions

hospitaldb

retail_store

sakila

socialmediadb

sys

Administration Schemas

Information

No object selected

SQL File 4 employeeDB session 5 joins class copy SQL File 7 session 5 joins reference

Limit to 1000 rows

```
322 Employee_Id,  
323 Employee_Name,  
324 Department_ID,  
325 Salary,  
326 sum(salary) over(partition by Department_ID order by Salary)  
327 As Running_Total_Salary  
328 From Employees;  
329
```

Result Grid

Employee_Id	Employee_Name	Department_ID	Salary	Running_Total_Salary
5020	Aarav Nair	1	90000.00	90000.00
5011	Saamvi Patel	2	60000.00	60000.00
5023	Anika Paul	2	70000.00	130000.00
5007	Anushka Singh	2	90000.00	220000.00
5001	Vihaan Singh	3	60000.00	60000.00
5026	Vihaan Singh	3	75000.00	135000.00
5025	Avin Iyengar	3	100000.00	235000.00
5019	Ishika Patel	4	55000.00	55000.00
5022	Adhva Ivenoor	4	60000.00	115000.00

Result 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	16:52:51	Select Employee_Id, Employee_Name, Department_ID, Salary, dense_Rank() over(...	30 row(s) returned	0.000 sec / 0.000 sec
10	16:58:59	select Employee_Id, Employee_Name, Department_ID, Salary, sum(salary) overpart...	30 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

04:59 PM 01-08-2026